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T.Y.B.Sc. (Computer Science) Sem-VI

CS-369: Object Oriented Programming using Java - II

Practical Slip Solutions (All 30 Slips)

Topics Covered:

Multithreading | *Collections* | *JDBC* | *Swing* | *Servlet* | *JSP* | *Spring*

Slip 1

Q1: Multithreading - Display alphabets A to Z every 2 seconds

```
// AlphabetThread.java
class AlphabetThread extends Thread {
    public void run() {
        for (char c = 'A'; c <= 'Z'; c++) {
            System.out.println(c);
            try {
                Thread.sleep(2000); // 2 seconds
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }
}

public class Slip1Q1 {
    public static void main(String[] args) {
        AlphabetThread t = new AlphabetThread();
        t.start();
    }
}
```

Q2: Employee JDBC with Swing (Eno, EName, Designation, Salary)

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class Slip1Q2 extends JFrame implements ActionListener {
    JTextField tEno, tEName, tDesig, tSalary;
    JButton btnSave;

    Slip1Q2() {
        setTitle("Employee Entry");
        setLayout(new GridLayout(5, 2, 5, 5));
        add(new JLabel("Eno:")); tEno = new JTextField(); add(tEno);
        add(new JLabel("EName:")); tEName = new JTextField(); add(tEName);
        add(new JLabel("Designation:")); tDesig = new JTextField(); add(tDesig);
        add(new JLabel("Salary:")); tSalary = new JTextField(); add(tSalary);
        btnSave = new JButton("Save"); add(btnSave);
        btnSave.addActionListener(this);
        setSize(400, 200); setVisible(true); setDefaultCloseOperation(EXIT_ON_CLOSE);
    }

    public void actionPerformed(ActionEvent e) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost/testdb", "root", "password");
            PreparedStatement ps = con.prepareStatement(
                "INSERT INTO Employee VALUES(?,?,?,?)");
            ps.setInt(1, Integer.parseInt(tEno.getText()));
            ps.setString(2, tEName.getText());
            ps.setString(3, tDesig.getText());
            ps.setDouble(4, Double.parseDouble(tSalary.getText()));
            ps.executeUpdate();
            JOptionPane.showMessageDialog(this, "Record Saved!");
            con.close();
        } catch (Exception ex) { JOptionPane.showMessageDialog(this, ex.getMessage()); }
    }
```

```
    public static void main(String[] args) { new Slip1Q2(); }
}
```

Slip 2

Q1: Read N friend names, store in HashSet, display in ascending order

```
import java.util.*;

public class Slip2Q1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: ");
        int n = sc.nextInt();
        HashSet<String> hs = new HashSet<>();
        for (int i = 0; i < n; i++) {
            System.out.print("Enter name: ");
            hs.add(sc.next());
        }
        TreeSet<String> ts = new TreeSet<>(hs); // sorted
        System.out.println("Names in ascending order: " + ts);
    }
}
```

Q2: Servlet - HTTP Request Info (IP, browser, OS, loaded servlets)

```
// InfoServlet.java
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class InfoServlet extends HttpServlet {
    public void doGet(HttpServletRequest req, HttpServletResponse res)
                        throws IOException, ServletException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        out.println("<h2>Request Info</h2>");
        out.println("IP: " + req.getRemoteAddr() + "<br>");
        out.println("Browser: " + req.getHeader("User-Agent") + "<br>");

        out.println("<h2>Server Info</h2>");
        ServletContext ctx = getServletContext();
        out.println("OS: " + System.getProperty("os.name") + "<br>");
        // List loaded servlets
        java.util.Enumeration<String> names = ctx.getServletNames();
        out.println("Loaded Servlets:<br>");
        while (names.hasMoreElements())
            out.println(names.nextElement() + "<br>");
        out.close();
    }
}
```

Slip 3

Q1: JSP - Display Patient details in tabular form

```
<!-- patient.jsp -->
<%@ page language="java" %>
<html><body>
<h2>Patient Details</h2>
<table border='1'>
  <tr><th>PNo</th><th>PName</th><th>Address</th><th>Age</th><th>Disease</th></tr>
  <%
    String[][] patients = {
      {"1","Ramesh","Pune","30","Fever"},
      {"2","Suresh","Mumbai","45","Diabetes"},
      {"3","Meena","Nashik","25","Cough"}
    };
    for(String[] p : patients) {
  %>
  <tr>
    <td><%=p[0]%></td><td><%=p[1]%></td>
    <td><%=p[2]%></td><td><%=p[3]%></td><td><%=p[4]%></td>
  </tr>
  <% } %>
</table>
</body></html>
```

Q2: LinkedList - add at end, delete first, display in reverse

```
import java.util.*;

public class Slip3Q2 {
    public static void main(String[] args) {
        LinkedList<String> ll = new LinkedList<>();
        // Add elements at end
        ll.addLast("Java");
        ll.addLast("Python");
        ll.addLast("C++");
        ll.addLast("HTML");
        System.out.println("After adding: " + ll);

        // Delete first element
        ll.removeFirst();
        System.out.println("After deleting first: " + ll);

        // Display in reverse order
        System.out.print("Reverse order: ");
        Iterator<String> it = ll.descendingIterator();
        while (it.hasNext()) System.out.print(it.next() + " ");
    }
}
```

Slip 4

Q1: Runnable interface - Blink Text on JFrame

```
import javax.swing.*;
import java.awt.*;

public class Slip4Q1 extends JFrame implements Runnable {
    JLabel label;
    boolean visible = true;

    Slip4Q1() {
        label = new JLabel("BLINKING TEXT", SwingConstants.CENTER);
        label.setFont(new Font("Arial", Font.BOLD, 30));
        add(label);
        setSize(400, 200);
        setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        new Thread(this).start();
    }

    public void run() {
        while (true) {
            label.setVisible(visible = !visible);
            try { Thread.sleep(500); } catch (Exception e) {}
        }
    }

    public static void main(String[] args) { new Slip4Q1(); }
}
```

Q2: City names and STD codes using HashMap (add, remove, search)

```
import java.util.*;

public class Slip4Q2 {
    public static void main(String[] args) {
        HashMap<String, Integer> map = new HashMap<>();
        // Add cities (no duplicates - HashMap handles this)
        map.put("Pune", 20);
        map.put("Mumbai", 22);
        map.put("Nashik", 253);
        map.put("Pune", 20); // duplicate - won't be added
        System.out.println("Cities: " + map);

        // Remove a city
        map.remove("Nashik");
        System.out.println("After remove: " + map);

        // Search
        String search = "Mumbai";
        if (map.containsKey(search))
            System.out.println(search + " STD: " + map.get(search));
        else
            System.out.println("City not found");
    }
}
```

Slip 5

Q1: Hashtable - mobile number & student name, display using Enumeration

```java
import java.util.*;

public class Slip5Q1 {
    public static void main(String[] args) {
        Hashtable<String, String> ht = new Hashtable<>();
        ht.put("9876543210", "Rahul");
        ht.put("9123456789", "Priya");
        ht.put("9988776655", "Suresh");

        System.out.println("Student Details:");
        Enumeration<String> keys = ht.keys();
        while (keys.hasMoreElements()) {
            String mob = keys.nextElement();
            System.out.println("Mobile: " + mob + " -> Name: " + ht.get(mob));
        }
    }
}
```

Q2: JSP - Online MCQ Test with random questions, score display

```jsp
<!-- quiz.jsp -->
<%@ page language="java" session="true" %>
<html><body>
<%
  String[][] questions = {
    {"What is JVM?","Java Virtual Machine","Java Visual Machine",
     "Java Variable Method","None","A"},
    {"Java is ___?","Compiled","Interpreted","Both","None","C"},
    {"OOP stands for?","Object Oriented Programming","Open Order Protocol",
     "Both","None","A"}
  };
  String ans = request.getParameter("answer");
  String qStr = request.getParameter("q");
  int q = (qStr == null) ? 0 : Integer.parseInt(qStr);
  int score = 0;
  String scoreStr = request.getParameter("score");
  if(scoreStr != null) score = Integer.parseInt(scoreStr);
  if(ans != null && qStr != null) {
    if(ans.equals(questions[q-1][5])) score++;
  }
  if(q >= questions.length) {
    out.println("<h2>Your Score: " + score + "/" + questions.length + "</h2>");
  } else {
    String[] row = questions[q];
%>
<form action='quiz.jsp' method='post'>
  <h3><%=row[0]%></h3>
  <input type='radio' name='answer' value='A'> <%=row[1]%><br>
  <input type='radio' name='answer' value='B'> <%=row[2]%><br>
  <input type='radio' name='answer' value='C'> <%=row[3]%><br>
  <input type='radio' name='answer' value='D'> <%=row[4]%><br>
  <input type='hidden' name='q' value='<%=q+1%>'>
  <input type='hidden' name='score' value='<%=score%>'>
  <input type='submit' value='Next'>
</form>
<% } %>
</body></html>
```

Slip 6

Q1: N integers in Collection (TreeSet), sorted, no duplicates, search

```
import java.util.*;

public class Slip6Q1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: ");
        int n = sc.nextInt();
        TreeSet<Integer> ts = new TreeSet<>();
        for (int i = 0; i < n; i++) {
            System.out.print("Enter integer: ");
            ts.add(sc.nextInt());
        }
        System.out.println("Sorted (no duplicates): " + ts);

        System.out.print("Enter element to search: ");
        int key = sc.nextInt();
        System.out.println(ts.contains(key) ? "Found" : "Not Found");
    }
}
```

Q2: Multithreading - Traffic Signal Simulation (Swing)

```
import javax.swing.*;
import java.awt.*;

public class Slip6Q2 extends JFrame implements Runnable {
    String color = "RED";
    JLabel lbl;

    Slip6Q2() {
        lbl = new JLabel("", SwingConstants.CENTER);
        lbl.setFont(new Font("Arial", Font.BOLD, 40));
        add(lbl);
        setSize(300, 200); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        new Thread(this).start();
    }

    public void run() {
        String[] colors = {"RED", "GREEN", "YELLOW"};
        Color[] clrs = {Color.RED, Color.GREEN, Color.YELLOW};
        int[] delays = {3000, 3000, 1000};
        int i = 0;
        while (true) {
            lbl.setText(colors[i]);
            lbl.setForeground(clrs[i]);
            try { Thread.sleep(delays[i]); } catch (Exception e) {}
            i = (i + 1) % 3;
        }
    }

    public static void main(String[] args) { new Slip6Q2(); }
}
```

Slip 7

Q1: Multi-thread - 3 threads: random number, if even square, if odd cube

```
public class Slip7Q1 {
    static int number;

    public static void main(String[] args) throws Exception {
        Thread generator = new Thread(() -> {
            while (true) {
                number = (int)(Math.random() * 100);
                System.out.println("Generated: " + number);
                try { Thread.sleep(1000); } catch (Exception e) {}
            }
        });

        Thread squareThread = new Thread(() -> {
            while (true) {
                if (number % 2 == 0)
                    System.out.println("Square of " + number + " = " + (number *
number));
                try { Thread.sleep(1000); } catch (Exception e) {}
            }
        });

        Thread cubeThread = new Thread(() -> {
            while (true) {
                if (number % 2 != 0)
                    System.out.println("Cube of " + number + " = " +
(number*number*number));
                try { Thread.sleep(1000); } catch (Exception e) {}
            }
        });

        generator.start();
        Thread.sleep(100);
        squareThread.start();
        cubeThread.start();
    }
}
```

Q2: JDBC - Create Product table, insert 5 records, display all

```
import java.sql.*;

public class Slip7Q2 {
    public static void main(String[] args) throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost/testdb", "root", "password");
        Statement st = con.createStatement();

        // Create table
        st.execute("CREATE TABLE IF NOT EXISTS Product(" +
            "Pid INT PRIMARY KEY, Pname VARCHAR(50), Price DOUBLE)");

        // Insert 5 records
        st.execute("INSERT INTO Product VALUES(1,'Laptop',55000)");
        st.execute("INSERT INTO Product VALUES(2,'Mouse',500)");
        st.execute("INSERT INTO Product VALUES(3,'Keyboard',800)");
        st.execute("INSERT INTO Product VALUES(4,'Monitor',12000)");
        st.execute("INSERT INTO Product VALUES(5,'Printer',8000)");

        // Display all records
```

```
        ResultSet rs = st.executeQuery("SELECT * FROM Product");
        System.out.println("Pid\tPname\t\tPrice");
        while (rs.next())
            System.out.println(rs.getInt(1)+"\t"+rs.getString(2)+"\t\t"+rs.getDouble(3));
        con.close();
    }
}
```

Slip 8

Q1: Thread - print text n times (3 threads: COVID19, LOCKDOWN2020, VACCINATED2021)

```
class PrintThread extends Thread {
    String text;
    int times;

    PrintThread(String text, int times) {
        this.text = text;
        this.times = times;
    }

    public void run() {
        for (int i = 1; i <= times; i++)
            System.out.println(getName() + ": " + text + " (" + i + ")");
    }
}

public class Slip8Q1 {
    public static void main(String[] args) {
        new PrintThread("COVID19", 10).start();
        new PrintThread("LOCKDOWN2020", 20).start();
        new PrintThread("VACCINATED2021", 30).start();
    }
}
```

Q2: JSP - Check prime number, display result in red color

```
<!-- prime.jsp -->
<%@ page language="java" %>
<html><body>
<form method='post'>
  Enter Number: <input type='text' name='num'>
  <input type='submit' value='Check'>
</form>
<%
  String numStr = request.getParameter("num");
  if(numStr != null && !numStr.isEmpty()) {
    int n = Integer.parseInt(numStr);
    boolean prime = n > 1;
    for(int i = 2; i <= Math.sqrt(n); i++) {
      if(n % i == 0) { prime = false; break; }
    }
%>
  <font color='red'><b>
    <%=n%> is <%=prime ? "Prime" : "Not Prime"%>
  </b></font>
<% } %>
</body></html>
```

Slip 9

Q1: Thread - Moving ball in panel vertically on Start button click (Swing)

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Slip9Q1 extends JFrame {
    int y = 20;
    JPanel panel;

    Slip9Q1() {
        JButton btn = new JButton("Start");
        panel = new JPanel() {
            public void paintComponent(Graphics g) {
                super.paintComponent(g);
                g.setColor(Color.RED);
                g.fillOval(150, y, 30, 30);
            }
        };
        panel.setPreferredSize(new Dimension(350, 400));
        btn.addActionListener(e -> {
            new Thread(() -> {
                while (true) {
                    y += 5;
                    if (y > 370) y = 20;
                    panel.repaint();
                    try { Thread.sleep(50); } catch (Exception ex) {}
                }
            }).start();
        });
        add(panel, BorderLayout.CENTER);
        add(btn, BorderLayout.SOUTH);
        pack(); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
    }

    public static void main(String[] args) { new Slip9Q1(); }
}
```

Q2: Spring - Display a message

```
// HelloBean.java
public class HelloBean {
    private String message;
    public void setMessage(String msg) { this.message = msg; }
    public String getMessage() { return message; }
    public void displayMessage() { System.out.println(message); }
}

// applicationContext.xml
// <bean id='helloBean' class='HelloBean'>
//   <property name='message' value="If you can't explain it simply,
//     you don't understand it well enough"/>
// </bean>

// MainApp.java
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class MainApp {
    public static void main(String[] args) {
        ApplicationContext ctx =
```

```
            new ClassPathXmlApplicationContext("applicationContext.xml");
        HelloBean bean = (HelloBean) ctx.getBean("helloBean");
        bean.displayMessage();
    }
}
```

Slip 10

Q1: Spring - Display Current Date

```
// DateBean.java
import java.util.Date;
public class DateBean {
    public String getCurrentDate() {
        return new Date().toString();
    }
}

// applicationContext.xml
// <bean id='dateBean' class='DateBean'/>

// MainApp.java
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class MainApp {
    public static void main(String[] args) {
        ApplicationContext ctx =
            new ClassPathXmlApplicationContext("applicationContext.xml");
        DateBean db = (DateBean) ctx.getBean("dateBean");
        System.out.println("Current Date: " + db.getCurrentDate());
    }
}
```

Q2: JDBC - Display first record from Student table on TextFields (Swing)

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class Slip10Q2 extends JFrame implements ActionListener {
    JTextField tRNo, tSName, tPer;
    JButton btn;

    Slip10Q2() {
        setLayout(new GridLayout(4, 2, 5, 5));
        add(new JLabel("RNo:")); tRNo = new JTextField(); add(tRNo);
        add(new JLabel("SName:")); tSName = new JTextField(); add(tSName);
        add(new JLabel("Per:")); tPer = new JTextField(); add(tPer);
        btn = new JButton("Fetch First Record"); add(btn);
        btn.addActionListener(this);
        setSize(400, 200); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
    }

    public void actionPerformed(ActionEvent e) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost/testdb", "root", "password");
            ResultSet rs = con.createStatement().executeQuery(
                "SELECT * FROM Student LIMIT 1");
            if (rs.next()) {
                tRNo.setText(String.valueOf(rs.getInt("RNo")));
                tSName.setText(rs.getString("SName"));
                tPer.setText(String.valueOf(rs.getDouble("Per")));
            }
            con.close();
        } catch (Exception ex) { JOptionPane.showMessageDialog(this, ex.getMessage()); }
```

```
    }

    public static void main(String[] args) { new Slip10Q2(); }
}
```

Slip 11

Q1: Servlet - Search customer by number in database

```
// search.html
<form action='SearchServlet' method='post'>
  Customer No: <input type='text' name='cno'>
  <input type='submit' value='Search'>
</form>

// SearchServlet.java
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
import java.sql.*;

public class SearchServlet extends HttpServlet {
    public void doPost(HttpServletRequest req, HttpServletResponse res)
                        throws IOException, ServletException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        int cno = Integer.parseInt(req.getParameter("cno"));
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost/testdb", "root", "password");
            PreparedStatement ps = con.prepareStatement(
                "SELECT * FROM Customer WHERE CNo=?");
            ps.setInt(1, cno);
            ResultSet rs = ps.executeQuery();
            if (rs.next())
                out.println("Name: "+rs.getString("CName")+" City:
"+rs.getString("City"));
            else
                out.println("Customer not found!");
            con.close();
        } catch (Exception ex) { out.println("Error: " + ex.getMessage()); }
    }
}
```

Q2: JDBC - Display column info of DONAR table using ResultSetMetaData

```
import java.sql.*;

public class Slip11Q2 {
    public static void main(String[] args) throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost/testdb", "root", "password");
        ResultSet rs = con.createStatement().executeQuery("SELECT * FROM DONAR");
        ResultSetMetaData rsmd = rs.getMetaData();
        int cols = rsmd.getColumnCount();
        System.out.println("Column Count: " + cols);
        for (int i = 1; i <= cols; i++) {
            System.out.println("Column " + i + ":");
            System.out.println("  Name: " + rsmd.getColumnName(i));
            System.out.println("  Type: " + rsmd.getColumnTypeName(i));
            System.out.println("  Size: " + rsmd.getColumnDisplaySize(i));
        }
        con.close();
    }
}
```

Slip 12

Q1: JSP - Check perfect number (Use Include directive)

```
<!-- perfect_check.jsp (included file) -->
<%@ page language="java" %>
<%
  int n = Integer.parseInt(request.getParameter("num"));
  int sum = 0;
  for(int i = 1; i < n; i++) if(n % i == 0) sum += i;
  String result = (sum == n) ? n + " is a Perfect Number" : n + " is NOT Perfect";
%>
<b><%=result%></b>

<!-- index.jsp (main file) -->
<%@ page language="java" %>
<html><body>
<form method='post'>
  Enter Number: <input type='text' name='num'>
  <input type='submit' value='Check'>
</form>
<%
  String n = request.getParameter("num");
  if(n != null && !n.isEmpty()) {
%>
  <%@ include file="perfect_check.jsp" %>
<% } %>
</body></html>
```

Q2: JDBC - PROJECT table (Swing JTable display)

```
import javax.swing.*;
import java.sql.*;

public class Slip12Q2 extends JFrame {
    Slip12Q2() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost/testdb", "root", "password");
            Statement st = con.createStatement();
            st.execute("CREATE TABLE IF NOT EXISTS PROJECT(" +
                "project_id INT PRIMARY KEY, Project_name VARCHAR(50)," +
                "Project_description VARCHAR(100), Project_Status VARCHAR(20))");
            st.execute("INSERT IGNORE INTO PROJECT VALUES(1,'ERP','Enterprise Resource
Planning','Active')");
            st.execute("INSERT IGNORE INTO PROJECT VALUES(2,'CRM','Customer Relationship
Mgmt','Done')");

            ResultSet rs = st.executeQuery("SELECT * FROM PROJECT");
            String[] cols = {"ID","Name","Description","Status"};
            java.util.Vector<Object[]> data = new java.util.Vector<>();
            while(rs.next())
                data.add(new
Object[]{rs.getInt(1),rs.getString(2),rs.getString(3),rs.getString(4)});
            Object[][] tableData = data.toArray(new Object[0][]);
            add(new JScrollPane(new JTable(tableData, cols)));
            con.close();
        } catch(Exception e) { e.printStackTrace(); }
        setSize(600, 300); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
    }
    public static void main(String[] args) { new Slip12Q2(); }
}
```

Slip 13

Q1: JDBC - Display database info and all tables (DatabaseMetaData)

```
import java.sql.*;

public class Slip13Q1 {
    public static void main(String[] args) throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost/testdb", "root", "password");
        DatabaseMetaData dmd = con.getMetaData();
        System.out.println("DB Product: " + dmd.getDatabaseProductName());
        System.out.println("DB Version: " + dmd.getDatabaseProductVersion());
        System.out.println("Driver: " + dmd.getDriverName());
        System.out.println("URL: " + dmd.getURL());
        System.out.println("User: " + dmd.getUserName());

        // List all tables
        ResultSet rs = dmd.getTables(null, null, "%", new String[]{"TABLE"});
        System.out.println("\nTables:");
        while (rs.next())
            System.out.println("  " + rs.getString("TABLE_NAME"));
        con.close();
    }
}
```

Q2: Thread lifecycle - creation, sleep, dead (random sleep 0-4999ms)

```
class LifeThread extends Thread {
    int sleepTime;

    LifeThread(String name) {
        super(name);
        sleepTime = (int)(Math.random() * 5000);
    }

    public void run() {
        System.out.println(getName() + " created, sleep: " + sleepTime + "ms");
        try {
            Thread.sleep(sleepTime);
        } catch (InterruptedException e) {}
        System.out.println(getName() + " is dead.");
    }
}

public class Slip13Q2 {
    public static void main(String[] args) {
        new LifeThread("Alpha").start();
        new LifeThread("Beta").start();
        new LifeThread("Gamma").start();
    }
}
```

Slip 14

Q1: Multithreading - Simple Search Engine (search string in text files)

```
import java.io.*;
import java.util.*;

class FileSearchThread extends Thread {
    String filename, keyword;

    FileSearchThread(String filename, String keyword) {
        this.filename = filename;
        this.keyword = keyword;
    }

    public void run() {
        try (BufferedReader br = new BufferedReader(new FileReader(filename))) {
            String line;
            int lineNo = 0;
            while ((line = br.readLine()) != null) {
                lineNo++;
                if (line.contains(keyword))
                    System.out.println("Found in " + filename + " at line " + lineNo);
            }
        } catch (IOException e) { System.out.println("Error: " + filename); }
    }
}

public class Slip14Q1 {
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter search string: ");
        String keyword = sc.nextLine();
        File dir = new File(".");
        for (File f : dir.listFiles((d, n) -> n.endsWith(".txt")))
            new FileSearchThread(f.getName(), keyword).start();
    }
}
```

Q2: JSP - Sum of first and last digit, display in Red color font size 18

```
<!-- sumdigit.jsp -->
<%@ page language="java" %>
<html><body>
<form method='post'>
  Enter Number: <input type='text' name='num'>
  <input type='submit' value='Calculate'>
</form>
<%
  String numStr = request.getParameter("num");
  if(numStr != null && !numStr.isEmpty()) {
    String s = numStr.replaceAll("-","");
    int first = s.charAt(0) - '0';
    int last = s.charAt(s.length()-1) - '0';
    int sum = first + last;
%>
  <font color='red' size='18'><b>
    Sum of first (<%=first%>) and last (<%=last%>) digit = <%=sum%>
  </b></font>
<% } %>
</body></html>
```

Slip 15

Q1: Thread - Display name and priority

```
class MyThread extends Thread {
    MyThread(String name, int priority) {
        setName(name);
        setPriority(priority);
    }
    public void run() {
        System.out.println("Thread Name: " + getName() +
            " | Priority: " + getPriority());
    }
}

public class Slip15Q1 {
    public static void main(String[] args) {
        new MyThread("LowPriority", Thread.MIN_PRIORITY).start();
        new MyThread("NormalPriority", Thread.NORM_PRIORITY).start();
        new MyThread("HighPriority", Thread.MAX_PRIORITY).start();
    }
}
```

Q2: Servlet - Visit counter using Cookie

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class VisitServlet extends HttpServlet {
    public void doGet(HttpServletRequest req, HttpServletResponse res)
                        throws IOException, ServletException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        Cookie[] cookies = req.getCookies();
        int count = 0;
        if (cookies != null) {
            for (Cookie c : cookies)
                if (c.getName().equals("visitCount"))
                    count = Integer.parseInt(c.getValue());
        }
        count++;
        res.addCookie(new Cookie("visitCount", String.valueOf(count)));
        if (count == 1)
            out.println("<h2>Welcome! First time visitor.</h2>");
        else
            out.println("<h2>Welcome back! You have visited " + count + " times.</h2>");
    }
}
```

Slip 16

Q1: TreeSet - Add colors, print in ascending order

```
import java.util.*;

public class Slip16Q1 {
    public static void main(String[] args) {
        TreeSet<String> ts = new TreeSet<>();
        ts.add("Red");
        ts.add("Blue");
        ts.add("Green");
        ts.add("Yellow");
        ts.add("White");
        ts.add("Black");
        System.out.println("Colors in ascending order: " + ts);
    }
}
```

Q2: JDBC - Teacher table (PreparedStatement) - display JAVA teachers

```
import java.sql.*;

public class Slip16Q2 {
    public static void main(String[] args) throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost/testdb", "root", "password");

        // Insert 5 records
        PreparedStatement ps = con.prepareStatement(
            "INSERT INTO Teacher(TNo,TName,Subject) VALUES(?,?,?)");
        String[][] data = {{"1","Ramesh","JAVA"},{"2","Suresh","C++"},
            {"3","Meena","JAVA"},{"4","Pooja","Python"},{"5","Amit","JAVA"}};
        for (String[] d : data) {
            ps.setInt(1, Integer.parseInt(d[0]));
            ps.setString(2, d[1]); ps.setString(3, d[2]);
            ps.executeUpdate();
        }

        // Display JAVA teachers
        PreparedStatement ps2 = con.prepareStatement(
            "SELECT * FROM Teacher WHERE Subject=?");
        ps2.setString(1, "JAVA");
        ResultSet rs = ps2.executeQuery();
        System.out.println("JAVA Teachers:");
        while (rs.next())
            System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getString(3));
        con.close();
    }
}
```

Slip 17

Q1: N integers, sorted, no duplicates (TreeSet)

```
import java.util.*;

public class Slip17Q1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: ");
        int n = sc.nextInt();
        TreeSet<Integer> ts = new TreeSet<>();
        for (int i = 0; i < n; i++) {
            System.out.print("Enter integer: ");
            ts.add(sc.nextInt());
        }
        System.out.println("Sorted (no duplicates): " + ts);
    }
}
```

Q2: Multithreading - Display 1 to 100 in JTextField on JButton click

```
import javax.swing.*;
import java.awt.*;

public class Slip17Q2 extends JFrame implements Runnable {
    JTextField tf;

    Slip17Q2() {
        tf = new JTextField();
        tf.setFont(new Font("Arial", Font.PLAIN, 20));
        JButton btn = new JButton("Start");
        btn.addActionListener(e -> new Thread(this).start());
        add(tf, BorderLayout.CENTER);
        add(btn, BorderLayout.SOUTH);
        setSize(300, 150); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
    }

    public void run() {
        for (int i = 1; i <= 100; i++) {
            final int num = i;
            SwingUtilities.invokeLater(() -> tf.setText(String.valueOf(num)));
            try { Thread.sleep(100); } catch (Exception e) {}
        }
    }

    public static void main(String[] args) { new Slip17Q2(); }
}
```

Slip 18

Q1: Multithreading - Display vowels from string every 3 seconds

```
public class Slip18Q1 {
    public static void main(String[] args) {
        String str = "Hello World Java Programming";
        new Thread(() -> {
            for (char c : str.toCharArray()) {
                if ("aeiouAEIOU".indexOf(c) != -1) {
                    System.out.println("Vowel: " + c);
                    try { Thread.sleep(3000); } catch (Exception e) {}
                }
            }
        }).start();
    }
}
```

Q2: Servlet - Student details (SeatNo, Name, Class, Marks) - Percentage & Grade

```
// student.html
<form action='GradeServlet' method='post'>
  SeatNo: <input name='sno'><br>
  Name: <input name='name'><br>
  Class: <input name='cls'><br>
  Total Marks: <input name='marks'><br>
  <input type='submit' value='Submit'>
</form>

// GradeServlet.java
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class GradeServlet extends HttpServlet {
    public void doPost(HttpServletRequest req, HttpServletResponse res)
                        throws IOException, ServletException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        String sno = req.getParameter("sno");
        String name = req.getParameter("name");
        String cls = req.getParameter("cls");
        double marks = Double.parseDouble(req.getParameter("marks"));
        double per = (marks / 600) * 100;
        String grade = per >= 75 ? "Distinction" : per >= 60 ? "First" :
                        per >= 50 ? "Second" : per >= 35 ? "Pass" : "Fail";
        out.println("<table border=1><tr><th>SeatNo</th><th>Name</th>" +
            "<th>Class</th><th>Marks</th><th>%</th><th>Grade</th></tr>");
        out.println("<tr><td>"+sno+"</td><td>"+name+"</td><td>"+cls+"</td>" +
            "<td>"+marks+"</td><td>"+per+"</td><td>"+grade+"</td></tr></table>");
    }
}
```

Slip 19

Q1: LinkedList - Accept N integers, display only negative integers

```
import java.util.*;

public class Slip19Q1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: ");
        int n = sc.nextInt();
        LinkedList<Integer> ll = new LinkedList<>();
        for (int i = 0; i < n; i++) {
            System.out.print("Enter integer: ");
            ll.add(sc.nextInt());
        }
        System.out.print("Negative integers: ");
        for (int x : ll)
            if (x < 0) System.out.print(x + " ");
    }
}
```

Q2: Servlet - Username/Password login with database validation

```
// LoginServlet.java
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;
import java.sql.*;

public class LoginServlet extends HttpServlet {
    public void doPost(HttpServletRequest req, HttpServletResponse res)
                        throws IOException, ServletException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        String user = req.getParameter("username");
        String pass = req.getParameter("password");
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost/testdb", "root", "password");
            PreparedStatement ps = con.prepareStatement(
                "SELECT * FROM users WHERE username=? AND password=?");
            ps.setString(1, user); ps.setString(2, pass);
            ResultSet rs = ps.executeQuery();
            if (rs.next())
                out.println("<h2>Welcome, " + user + "!</h2>");
            else
                out.println("<h2 style='color:red'>Invalid credentials!</h2>");
            con.close();
        } catch (Exception e) { out.println(e.getMessage()); }
    }
}
```

Slip 20

Q1: JSP - Accept number, display in words (e.g. 123 -> One Two Three)

```
<!-- numwords.jsp -->
<%@ page language="java" %>
<html><body>
<form method='post'>
  Enter Number: <input type='text' name='num'>
  <input type='submit' value='Convert'>
</form>
<%
  String numStr = request.getParameter("num");
  if(numStr != null && !numStr.isEmpty()) {
    String[] words = {"Zero","One","Two","Three","Four",
                      "Five","Six","Seven","Eight","Nine"};
    StringBuilder sb = new StringBuilder();
    for(char c : numStr.toCharArray()) {
      if(Character.isDigit(c))
        sb.append(words[c - '0']).append(" ");
    }
%>
  <font color='red' size='18'><b><%=sb.toString().trim()%></b></font>
<% } %>
</body></html>
```

Q2: Multithreading - Draw Temple using Swing

```
import javax.swing.*;
import java.awt.*;

public class Slip20Q2 extends JFrame implements Runnable {
    int step = 0;

    Slip20Q2() {
        setSize(400, 400); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        new Thread(this).start();
    }

    public void paint(Graphics g) {
        super.paint(g);
        g.setColor(Color.ORANGE);
        // Base
        if(step >= 1) g.fillRect(100, 280, 200, 80);
        // Pillars
        if(step >= 2) {
            g.setColor(Color.YELLOW);
            g.fillRect(120, 200, 20, 80);
            g.fillRect(260, 200, 20, 80);
        }
        // Roof
        if(step >= 3) {
            g.setColor(Color.RED);
            int[] xp = {80, 200, 320};
            int[] yp = {200, 120, 200};
            g.fillPolygon(xp, yp, 3);
        }
    }

    public void run() {
        for (step = 1; step <= 3; step++) {
            repaint();
            try { Thread.sleep(800); } catch (Exception e) {}
```

```
        }
    }

    public static void main(String[] args) { new Slip20Q2(); }
}
```

Slip 21

Q1: LinkedList - N subject names, display using Iterator

```
import java.util.*;

public class Slip21Q1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter N: ");
        int n = sc.nextInt();
        sc.nextLine();
        LinkedList<String> ll = new LinkedList<>();
        for (int i = 0; i < n; i++) {
            System.out.print("Enter subject name: ");
            ll.add(sc.nextLine());
        }
        System.out.println("Subjects:");
        Iterator<String> it = ll.iterator();
        while (it.hasNext())
            System.out.println(it.next());
    }
}
```

Q2: Multithreading - Producer Consumer problem (synchronization)

```
class SharedResource {
    int value;
    boolean available = false;

    synchronized void produce(int v) throws InterruptedException {
        while (available) wait();
        value = v;
        available = true;
        System.out.println("Produced: " + value);
        notifyAll();
    }

    synchronized int consume() throws InterruptedException {
        while (!available) wait();
        available = false;
        System.out.println("Consumed: " + value);
        notifyAll();
        return value;
    }
}

public class Slip21Q2 {
    public static void main(String[] args) {
        SharedResource sr = new SharedResource();
        new Thread(() -> {
            for(int i=1; i<=5; i++)
                try { sr.produce(i); Thread.sleep(500); } catch(Exception e){}
        }).start();
        new Thread(() -> {
            for(int i=1; i<=5; i++)
                try { sr.consume(); Thread.sleep(500); } catch(Exception e){}
        }).start();
    }
}
```

Slip 22

Q1: JDBC - Menu Driven Employee (Insert, Update, Display, Exit)

```
import java.sql.*;
import java.util.*;

public class Slip22Q1 {
    static Connection con;
    static Scanner sc = new Scanner(System.in);

    public static void main(String[] args) throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        con = DriverManager.getConnection(
            "jdbc:mysql://localhost/testdb", "root", "password");
        int ch;
        do {
            System.out.println("1.Insert 2.Update 3.Display 4.Exit");
            System.out.print("Choice: "); ch = sc.nextInt();
            switch(ch) {
                case 1: insert(); break;
                case 2: update(); break;
                case 3: display(); break;
            }
        } while(ch != 4);
        con.close();
    }

    static void insert() throws Exception {
        System.out.print("ENo EName Salary: ");
        PreparedStatement ps = con.prepareStatement(
            "INSERT INTO Employee VALUES(?,?,?)");
        ps.setInt(1, sc.nextInt());
        ps.setString(2, sc.next());
        ps.setDouble(3, sc.nextDouble());
        ps.executeUpdate(); System.out.println("Inserted.");
    }

    static void update() throws Exception {
        System.out.print("Enter ENo to update, new Salary: ");
        PreparedStatement ps = con.prepareStatement(
            "UPDATE Employee SET Salary=? WHERE ENo=?");
        ps.setDouble(1, sc.nextDouble());
        ps.setInt(2, sc.nextInt());
        ps.executeUpdate(); System.out.println("Updated.");
    }

    static void display() throws Exception {
        ResultSet rs = con.createStatement().executeQuery(
            "SELECT * FROM Employee");
        while(rs.next())
            System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getDouble(3));
    }
}
```

Q2: JSP - Accept Username, greet based on server time

```
<!-- greet.jsp -->
<%@ page language="java" import="java.util.*" %>
<html><body>
<form method='post'>
  UserName: <input type='text' name='uname'>
  <input type='submit' value='Greet'>
</form>
```

```
<%
  String uname = request.getParameter("uname");
  if(uname != null && !uname.isEmpty()) {
    int hour = Calendar.getInstance().get(Calendar.HOUR_OF_DAY);
    String greeting = (hour < 12) ? "Good Morning" :
                      (hour < 18) ? "Good Afternoon" : "Good Evening";
%>
  <h2><%=greeting%>, <%=uname%>!</h2>
<% } %>
</body></html>
```

Slip 23

Q1: Multithreading - Display each vowel from string every 3 seconds

```
import java.util.Scanner;

public class Slip23Q1 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str = sc.nextLine();
        new Thread(() -> {
            for (char c : str.toCharArray()) {
                if ("aeiouAEIOU".indexOf(c) != -1) {
                    System.out.println("Vowel: " + c);
                    try { Thread.sleep(3000); } catch (Exception e) {}
                }
            }
        }).start();
    }
}
```

Q2: N student names via command line - Iterator & ListIterator

```
import java.util.*;

public class Slip23Q2 {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        for (String name : args) list.add(name);

        System.out.println("Forward (Iterator):");
        Iterator<String> it = list.iterator();
        while (it.hasNext()) System.out.println(it.next());

        System.out.println("\nBackward (ListIterator):");
        ListIterator<String> lit = list.listIterator(list.size());
        while (lit.hasPrevious()) System.out.println(lit.previous());
    }
}
// Run: java Slip23Q2 Alice Bob Charlie
```

Slip 24

Q1: Multithreading - Scroll text left to right continuously (Swing)

```
import javax.swing.*;
import java.awt.*;

public class Slip24Q1 extends JFrame implements Runnable {
    int x = 0;
    JPanel panel;

    Slip24Q1() {
        panel = new JPanel() {
            public void paintComponent(Graphics g) {
                super.paintComponent(g);
                g.setFont(new Font("Arial", Font.BOLD, 24));
                g.setColor(Color.BLUE);
                g.drawString("Welcome to Java!", x, 50);
            }
        };
        add(panel);
        setSize(500, 120); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        new Thread(this).start();
    }

    public void run() {
        while (true) {
            x = (x + 5) % (getWidth() + 200);
            panel.repaint();
            try { Thread.sleep(50); } catch (Exception e) {}
        }
    }

    public static void main(String[] args) { new Slip24Q1(); }
}
```

Q2: JSP - Login page (username==password -> Login Successfully)

```
<!-- login.html -->
<form action='check.jsp' method='post'>
  Username: <input type='text' name='uname'><br>
  Password: <input type='password' name='pwd'><br>
  <input type='submit' value='Login'>
</form>

<!-- check.jsp -->
<%@ page language="java" %>
<%
  String u = request.getParameter("uname");
  String p = request.getParameter("pwd");
  if(u != null && u.equals(p)) {
    response.sendRedirect("Login.html");
  } else {
    response.sendRedirect("Error.html");
  }
%>

<!-- Login.html -->
<h2 style='color:green'>Login Successfully!</h2>

<!-- Error.html -->
<h2 style='color:red'>Login Failed!</h2>
```

Slip 25

Q1: JSP - Voter eligibility check (Name & Age)

```
<!-- voter.jsp -->
<%@ page language="java" %>
<html><body>
<form method='post'>
  Name: <input type='text' name='vname'><br>
  Age: <input type='text' name='age'><br>
  <input type='submit' value='Check'>
</form>
<%
  String vname = request.getParameter("vname");
  String ageStr = request.getParameter("age");
  if(vname != null && ageStr != null && !ageStr.isEmpty()) {
    int age = Integer.parseInt(ageStr);
    if(age >= 18)
      out.println("<h3>" + vname + " is ELIGIBLE to vote.</h3>");
    else
      out.println("<h3>" + vname + " is NOT eligible (age < 18).</h3>");
  }
%>
</body></html>
```

Q2: JDBC - DDL Query GUI (Create Table, Alter Table, Drop Table)

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class Slip25Q2 extends JFrame implements ActionListener {
    JTextField tQuery;
    JButton btnCreate, btnAlter, btnDrop;
    Connection con;

    Slip25Q2() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        con = DriverManager.getConnection(
            "jdbc:mysql://localhost/testdb", "root", "password");
        setLayout(new FlowLayout());
        add(new JLabel("Type Your DDL Query Here:"));
        tQuery = new JTextField(20); add(tQuery);
        btnCreate = new JButton("Create Table"); add(btnCreate);
        btnAlter = new JButton("Alter Table"); add(btnAlter);
        btnDrop = new JButton("Drop Table"); add(btnDrop);
        btnCreate.addActionListener(this);
        btnAlter.addActionListener(this);
        btnDrop.addActionListener(this);
        setSize(400, 200); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
    }

    public void actionPerformed(ActionEvent e) {
        try {
            con.createStatement().execute(tQuery.getText());
            JOptionPane.showMessageDialog(this, "Query Executed Successfully!");
        } catch (Exception ex) { JOptionPane.showMessageDialog(this, ex.getMessage()); }
    }

    public static void main(String[] args) throws Exception { new Slip25Q2(); }
}
```

Slip 26

Q1: JDBC - Delete employee by ID (PreparedStatement)

```
import java.sql.*;

public class Slip26Q1 {
    public static void main(String[] args) throws Exception {
        if (args.length == 0) {
            System.out.println("Usage: java Slip26Q1 <EmpID>"); return;
        }
        int eid = Integer.parseInt(args[0]);
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost/testdb", "root", "password");
        PreparedStatement ps = con.prepareStatement(
            "DELETE FROM Employee WHERE ENo=?");
        ps.setInt(1, eid);
        int rows = ps.executeUpdate();
        if (rows > 0) System.out.println("Employee deleted successfully.");
        else System.out.println("Employee ID not found.");
        con.close();
    }
}
```

Q2: JSP - Sum of first and last digit in Red color, font size 18

Note: Same as Slip 14 Q2 - refer to that solution above.

Slip 27

Q1: JDBC - Display College details on JTable (Swing)

```
import javax.swing.*;
import java.sql.*;

public class Slip27Q1 extends JFrame {
    Slip27Q1() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost/testdb", "root", "password");
        ResultSet rs = con.createStatement().executeQuery(
            "SELECT * FROM College");
        String[] cols = {"CID", "CName", "Address", "Year"};
        java.util.Vector<Object[]> rows = new java.util.Vector<>();
        while (rs.next())
            rows.add(new Object[]{rs.getInt(1), rs.getString(2),
                rs.getString(3), rs.getInt(4)});
        add(new JScrollPane(new JTable(rows.toArray(new Object[0][]), cols)));
        con.close();
        setSize(600, 300); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
    }
    public static void main(String[] args) throws Exception { new Slip27Q1(); }
}
```

Q2: Servlet - Change inactive session time interval

```
import javax.servlet.*;
import javax.servlet.http.*;
import java.io.*;

public class SessionServlet extends HttpServlet {
    public void doGet(HttpServletRequest req, HttpServletResponse res)
                        throws IOException, ServletException {
        res.setContentType("text/html");
        PrintWriter out = res.getWriter();
        HttpSession session = req.getSession();
        // Set max inactive interval to 60 seconds
        session.setMaxInactiveInterval(60);
        out.println("<h2>Session timeout set to " +
            session.getMaxInactiveInterval() + " seconds.</h2>");
        out.println("Session ID: " + session.getId());
    }
}
```

Slip 28

Q1: JSP - Accept string, display in reverse order

```
<!-- reverse.jsp -->
<%@ page language="java" %>
<html><body>
<form method='post'>
  Enter String: <input type='text' name='str'>
  <input type='submit' value='Reverse'>
</form>
<%
  String s = request.getParameter("str");
  if(s != null && !s.isEmpty()) {
    String rev = new StringBuilder(s).reverse().toString();
%>
  <h3>Reversed: <%=rev%></h3>
<% } %>
</body></html>
```

Q2: Display name of currently executing thread

```
class ShowThread extends Thread {
    ShowThread(String name) { super(name); }
    public void run() {
        System.out.println("Running: " + Thread.currentThread().getName());
        try { Thread.sleep(500); } catch (Exception e) {}
        System.out.println("Done: " + Thread.currentThread().getName());
    }
}

public class Slip28Q2 {
    public static void main(String[] args) {
        new ShowThread("Thread-Alpha").start();
        new ShowThread("Thread-Beta").start();
        new ShowThread("Thread-Gamma").start();
    }
}
```

Slip 29

Q1: JDBC - Display DONAR table column info using ResultSetMetaData

Note: Same as Slip 11 Q2 - refer to that solution above.

Q2: LinkedList of integers - add at first, delete last, display size

```
import java.util.*;

public class Slip29Q2 {
    public static void main(String[] args) {
        LinkedList<Integer> ll = new LinkedList<>();
        // Add at first position
        ll.addFirst(30);
        ll.addFirst(20);
        ll.addFirst(10);
        ll.addFirst(5);
        System.out.println("After adding at first: " + ll);

        // Delete last element
        ll.removeLast();
        System.out.println("After deleting last: " + ll);

        // Display size
        System.out.println("Size: " + ll.size());
    }
}
```

Slip 30

Q1: Multithreading - Draw Indian Flag (Swing)

```
import javax.swing.*;
import java.awt.*;

public class Slip30Q1 extends JFrame implements Runnable {
    int step = 0;

    Slip30Q1() {
        setSize(500, 350); setVisible(true);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        new Thread(this).start();
    }

    public void paint(Graphics g) {
        super.paint(g);
        int x = 100, y = 60, w = 280, h = 50;
        // Saffron
        if(step >= 1) { g.setColor(new Color(255, 153, 51)); g.fillRect(x, y, w, h); }
        // White
        if(step >= 2) { g.setColor(Color.WHITE); g.fillRect(x, y+h, w, h); }
        // Green
        if(step >= 3) { g.setColor(new Color(19, 136, 8)); g.fillRect(x, y+2*h, w, h); }
        // Ashoka Chakra
        if(step >= 4) {
            g.setColor(new Color(0, 0, 128));
            g.drawOval(x+w/2-20, y+h-20, 40, 40);
        }
        // Flagpole
        if(step >= 1) { g.setColor(Color.DARK_GRAY); g.fillRect(95, 55, 5, 200); }
    }

    public void run() {
        for(step = 1; step <= 4; step++) {
            repaint();
            try { Thread.sleep(600); } catch(Exception e) {}
        }
    }

    public static void main(String[] args) { new Slip30Q1(); }
}
```

Q2: JDBC - Scrollable ResultSet (Teacher table - TID, TName, Salary)

```
import java.sql.*;

public class Slip30Q2 {
    public static void main(String[] args) throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost/testdb", "root", "password");
        // TYPE_SCROLL_INSENSITIVE for scrollable ResultSet
        Statement st = con.createStatement(
            ResultSet.TYPE_SCROLL_INSENSITIVE,
            ResultSet.CONCUR_READ_ONLY);
        ResultSet rs = st.executeQuery("SELECT * FROM Teacher");

        System.out.println("--- Forward ---");
        while(rs.next())
            System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getDouble(3));

        System.out.println("--- Backward ---");
```

```
        rs.afterLast();
        while(rs.previous())
            System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getDouble(3));

        System.out.println("--- Last Record ---");
        rs.last();
        System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getDouble(3));
        con.close();
    }
}
```